



Graphs in Machine Learning

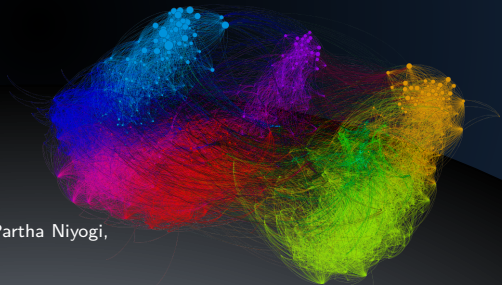
Laplacian SVMs and Max-Margin Graph Cuts

Inductive SSL Methods

Michal Valko

Isara Labs

Partially based on material by: Mikhail Belkin, Partha Niyogi,
Olivier Chapelle, Bernhard Schölkopf



SSL with Graphs: Laplacian SVMs

$$f^* = \arg \min_{f \in \mathcal{H}_{\mathcal{K}}} \sum_i^{n_l} \max(0, 1 - yf(\mathbf{x})) + \gamma_1 \|f\|_{\mathcal{K}}^2 + \gamma_2 \mathbf{f}^T \mathbf{L} \mathbf{f}$$

$\mathcal{H}_{\mathcal{K}}$ is nice and expressive.

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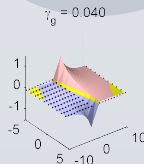
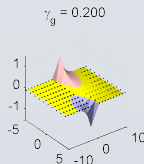
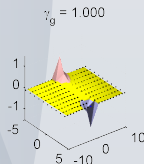
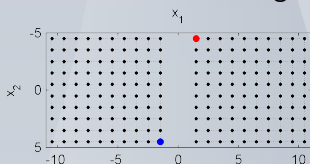
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Consider again this 2D data and linear $\mathcal{K}$.



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Linear $\mathcal{K} \equiv$ functions with slope α_1 and intercept α_2 .

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$$\min_{\alpha_1, \alpha_2} \sum_i^{n_l} V(f, \mathbf{x}_i, y_i) + \gamma_1$$

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$$\min_{\alpha_1, \alpha_2} \sum_i^{n_l} V(f, \mathbf{x}_i, y_i) + \gamma_1 [\alpha_1^2 + \alpha_2^2] + \gamma_2 \mathbf{f}^\top \mathbf{L} \mathbf{f}$$

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$$\mathbf{f}^\top \mathbf{L} \mathbf{f} = \frac{1}{2} \sum_{i,j} w_{ij} (f(\mathbf{x}_i) - f(\mathbf{x}_j))^2$$

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SSL with Graphs: Laplacian SVMs

2D data and linear $\mathcal{K}$ objective

$$\min_{\alpha_1, \alpha_2} \sum_i^{n_l} V(f, \mathbf{x}_i, y_i) + \left(\gamma_1 + \frac{\gamma_2 \Delta}{2} \right) [\alpha_1^2 + \alpha_2^2]$$

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The only influence of unlabeled data is through $\bar{\gamma}$.

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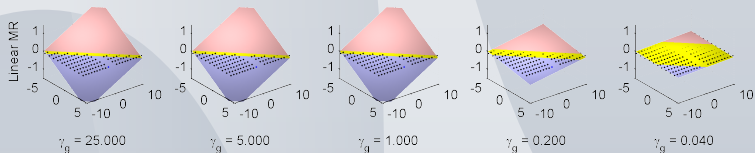
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The only influence of unlabeled data is through $\bar{\gamma}$.

The same value of the objective as for supervised learning for some γ **without the unlabeled data!** This is not good.

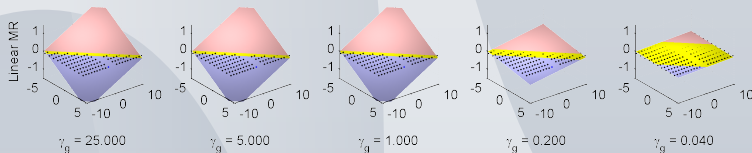
SSL with Graphs: Laplacian SVMs

LSVM for 2D data and linear $\mathcal{K}$ only changes the slope



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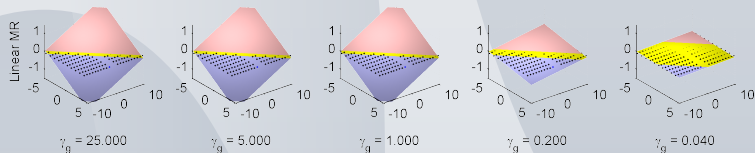
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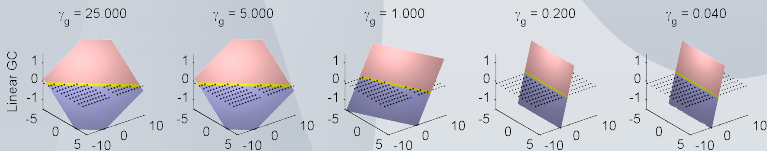
What would we like to see?

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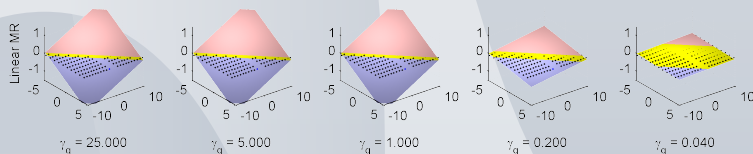


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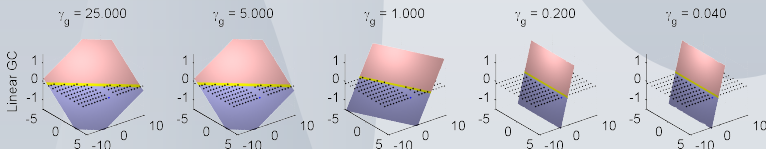


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One solution: We use the unlabeled data **before** optimizing over $\mathcal{H}_{\mathcal{K}}$!

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Let's take the confident data and use them as true!

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$$\begin{aligned} f^* &= \min_{f \in \mathcal{H}_{\mathcal{K}}} \sum_{i: |\ell_i^*| \geq \varepsilon} V(f, \mathbf{x}_i, \text{sgn}(\ell_i^*)) + \gamma \|f\|_{\mathcal{K}}^2 \\ \text{s.t. } \ell^* &= \arg \min_{\ell \in \mathbb{R}^N} \ell^\top (\mathbf{L} + \gamma_g \mathbf{I}) \ell \\ \text{s.t. } \ell_i &= y_i \text{ for all } i = 1, \dots, n_l \end{aligned}$$

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Representer theorem is still cool:

$$f^*(\mathbf{x}) = \sum_{i: |f_i^*| \geq \varepsilon} \alpha_i^* \mathcal{K}(\mathbf{x}_i, \mathbf{x})$$

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<https://misovalko.github.io/mva-ml-graphs.html>